

CHAPTER 04

SOFTWARE AND HARDWARE REQUIREMENTS

4.1 Software Requirements

- **Development Environment:** Mac OS

- **Code Editors:**

Visual Studio Code – A lightweight but powerful code editor with support for various programming languages and extensions.

- **Frontend Development Tools:**

HTML5: For structuring the web pages.

CSS3: For styling and responsive design.

TypeScript: Adds static typing to JavaScript, helping prevent bugs.

React: A JavaScript library for building user interfaces.

Vite: A fast build tool and development server for modern web projects.

- **Frontend Libraries and Frameworks:**

Recharts: A composable charting library built on React for creating visual data representations.

Lucide-react: For including icons.

Tailwind CSS: A utility-first CSS framework for styling.

ShadCN/UI – For UI components in React.

- **Backend Development Tools:**

Python: Used for backend logic, machine learning model integration, and API development.

Fast API: A modern, fast (high-performance) web framework for building APIs with Python.

Pickle: For loading the trained SVM model (svm_model.pkl).

NumPy & Pandas: For numerical and data manipulation tasks.

- **AI and ML Tools:**

OpenAI's LLM (e.g., GPT models): Used for generating responses, understanding inputs, or performing intelligent analysis.

Pinecone: A vector database used for efficient similarity search in AI tasks.

4.2 Hardware Requirements

- **Development Machine (for coding and testing):**

Processor: Apple M1 chip, macOS Sequoia.

RAM: Minimum 16 GB (32 GB recommended for faster processing during model operations and UI rendering).

Storage: Minimum 512 GB SSD (more if handling large datasets or models).

Display: Full HD (1920x1080) or higher for clear visualization during UI design and debugging



Fig no. 4.1 Python



Fig no. 4.2 Machine Learning



Fig no. 4.3 Vite



Fig no. 4.4 HTML and CSS